

STA 445 S26 Assignment 6

2026-04-15

Load your packages here:

Problem 1

A common task is to take a set of data that has multiple categorical variables and create a table of the number of cases for each combination. An introductory statistics textbook contains a data set summarizing student surveys from several sections of an intro class. The two variables of interest are **Gender** and **Year** which are the students gender and year in college. *Note: you will need to refer to Chapter 4 and Chapter 7 for some of the operations needed below - this is a great time to review chapter 4!*

a) Download the data set using the following:

I had to change this to just r instead of r=False because it wouldnt let me knit I hope you can understand this.

```
Survey <- read.csv('https://www.lock5stat.com/datasets2e/StudentSurvey.csv', na.strings=c('', ' '))
```

b) Select the specific columns of interest **Year** and **Gender**

```
Survey <-  
  Survey %>%  
  select(Year, Gender)  
head(Survey)
```

```
##      Year Gender  
## 1   Senior     M  
## 2 Sophomore    F  
## 3 FirstYear    M  
## 4   Junior     M  
## 5 Sophomore    F  
## 6 Sophomore    F
```

c) Convert the **Year** column to factors and properly order the factors based on common US progression (FirstYear - Sophomore - Junior - Senior)

```
Survey <-  
  Survey %>%  
  mutate(Year = fct_relevel(Year, "FirstYear", "Sophomore", "Junior", "Senior"))  
head(Survey)
```

```
##      Year Gender
## 1   Senior     M
## 2 Sophomore    F
## 3 FirstYear     M
## 4   Junior     M
## 5 Sophomore    F
## 6 Sophomore    F
```

```
levels(Survey$Year)
```

```
## [1] "FirstYear" "Sophomore" "Junior"      "Senior"
```

d) Convert the **Gender** column to factors and rename them Male/Female.

```
Survey <-
  Survey %>%
    mutate(Gender = factor(Gender)) %>%
    mutate(Gender = fct_recode(Gender, "Male" = "M", "Female" = "F"))
head(Survey)
```

```
##      Year Gender
## 1   Senior  Male
## 2 Sophomore Female
## 3 FirstYear  Male
## 4   Junior  Male
## 5 Sophomore Female
## 6 Sophomore Female
```

e) Produce a data set with eight rows and three columns that contains the number of responses for each gender:year combination. *You might want to look at the following functions: `dplyr::count` and `dplyr::drop_na`.*

```
Survey_count <-
  Survey %>%
    drop_na(Gender, Year) %>%
    count(Gender, Year)
Survey_count
```

```
##   Gender      Year  n
## 1 Female FirstYear 43
## 2 Female Sophomore 96
## 3 Female   Junior 18
## 4 Female   Senior 10
## 5  Male FirstYear 51
## 6  Male Sophomore 99
## 7  Male   Junior 17
## 8  Male   Senior 26
```

f) Pivot the table in part (e) to produce a table of the number of responses in the following form:

	Gender	First Year	Sophomore	Junior	Senior
	Female				
	Male				

```
Survey_table <-
  Survey_count %>%
    pivot_wider(names_from = Year, values_from = n, names_repair = "unique")
Survey_table
```

```
## # A tibble: 2 x 5
##   Gender FirstYear Sophomore Junior Senior
##   <fct>      <int>      <int>  <int>  <int>
## 1 Female         43         96     18     10
## 2 Male          51         99     17     26
```

Problem 2

For this problem we will consider two simple data sets. Remember to erase `eval = FALSE` from the code chunk below.

```
A <- tribble(
  ~Name, ~Car,
  'Alice', 'Ford F150',
  'Bob',   'Tesla Model III',
  'Charlie', 'VW Bug')

B <- tribble(
  ~First.Name, ~Pet,
  'Bob', 'Cat',
  'Charlie', 'Dog',
  'Alice', 'Rabbit')
```

- a. Squish the data frames together to generate a data set with three rows and three columns. Do two ways: first using `cbind` and then using one of the `dplyr` join commands.

```
cbind(A, B) %>%
  select(Name, Car, Pet)
```

```
##   Name      Car      Pet
## 1  Alice  Ford F150   Cat
## 2   Bob Tesla Model III Dog
## 3 Charlie   VW Bug  Rabbit
```

```
left_join(A, B, by = c("Name" = "First.Name"))
```

```
## # A tibble: 3 x 3
##   Name      Car      Pet
##   <chr>    <chr>    <chr>
## 1 Alice   Ford F150   Rabbit
## 2 Bob     Tesla Model III Cat
## 3 Charlie VW Bug     Dog
```

- b. It turns out that Alice also has a pet guinea pig. Add another row to the B data set. Do this using either the base function `rbind`, or either of the `dplyr` functions `add_row` or `bind_rows`.

```
B <- B %>%
  add_row(First.Name = "Alice", Pet = "Guinea Pig")
B
```

```
## # A tibble: 4 x 2
##   First.Name Pet
##   <chr>      <chr>
## 1 Bob       Cat
## 2 Charlie  Dog
## 3 Alice    Rabbit
## 4 Alice    Guinea Pig
```

- c. Squish the A and B data sets together to generate a data set with four rows and three columns. Do this two ways: first using `cbind` and then using one of the `dplyr` join commands. Which was easier to program? Which is more likely to have an error.

```
cbind(A, B) %>%
  select(Name, Car, Pet)
```

```
## Error in data.frame(..., check.names = FALSE): arguments imply differing number of rows: 3, 4
```

```
left_join(A, B, by = c("Name" = "First.Name"))
```

```
## # A tibble: 4 x 3
##   Name      Car      Pet
##   <chr>    <chr>    <chr>
## 1 Alice   Ford F150   Rabbit
## 2 Alice   Ford F150   Guinea Pig
## 3 Bob     Tesla Model III Cat
## 4 Charlie VW Bug     Dog
```

`left_join()` is easier to write because it matches rows by the same variable, whereas `cbind()` is easier to write incorrectly since it matches on position. Matching on position often creates problems for data frames with different numbers of rows.

Problem 3

From this book's GitHub there is a .csv file of the daily maximum temperature in Flagstaff at the Pulliam Airport. The link is: https://raw.githubusercontent.com/BuscagliaR/STA_444_v2/master/data-row/FlagMaxTemp.csv

```
temperature <- read_csv("https://raw.githubusercontent.com/BuscagliaR/STA_444_v2/master/data-row/FlagMaxTemp.csv")

## New names:
## Rows: 365 Columns: 34
## -- Column specification
## ----- Delimiter: "," dbl
## (34): ...1, Year, Month, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, ...
## i Use `spec()` to retrieve the full column specification for this data. i
## Specify the column types or set `show_col_types = FALSE` to quiet this message.
## * `` -> `...1`
```

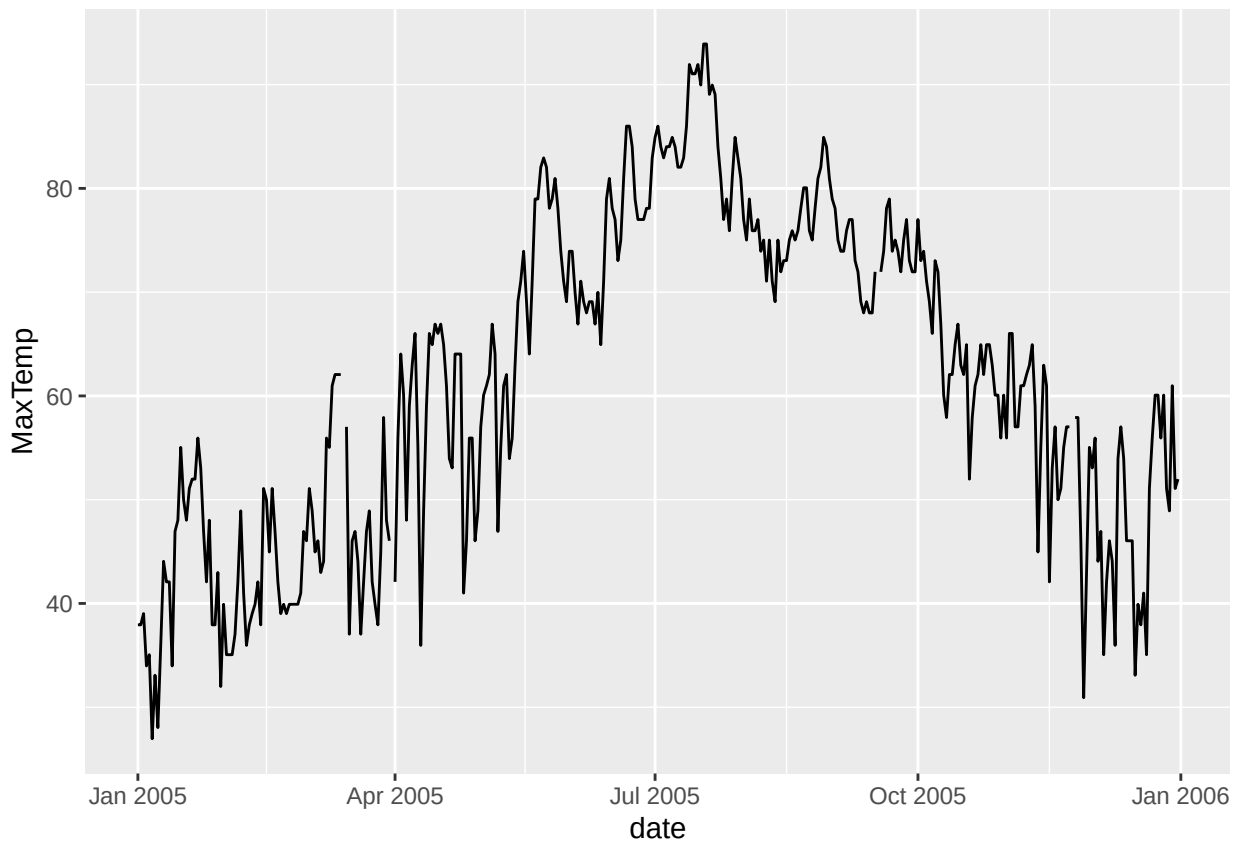
```
temperature_long <- temperature %>%
  pivot_longer(
    cols = -c(...1, Year, Month),
    names_to = "day",
    values_to = "MaxTemp"
  ) %>%
  mutate(day = as.numeric(day), date = make_date(Year, Month, day))
```

I needed to add the temperature long using the pivot longer so i can correctly format all of the dates. I hope you understand this.

a) Create a line graph that gives the daily maximum temperature for 2005. *Make sure the x-axis is a date and covers the whole year.*

```
temp_2005 <- temperature_long %>%
  filter(year(date) == 2005)

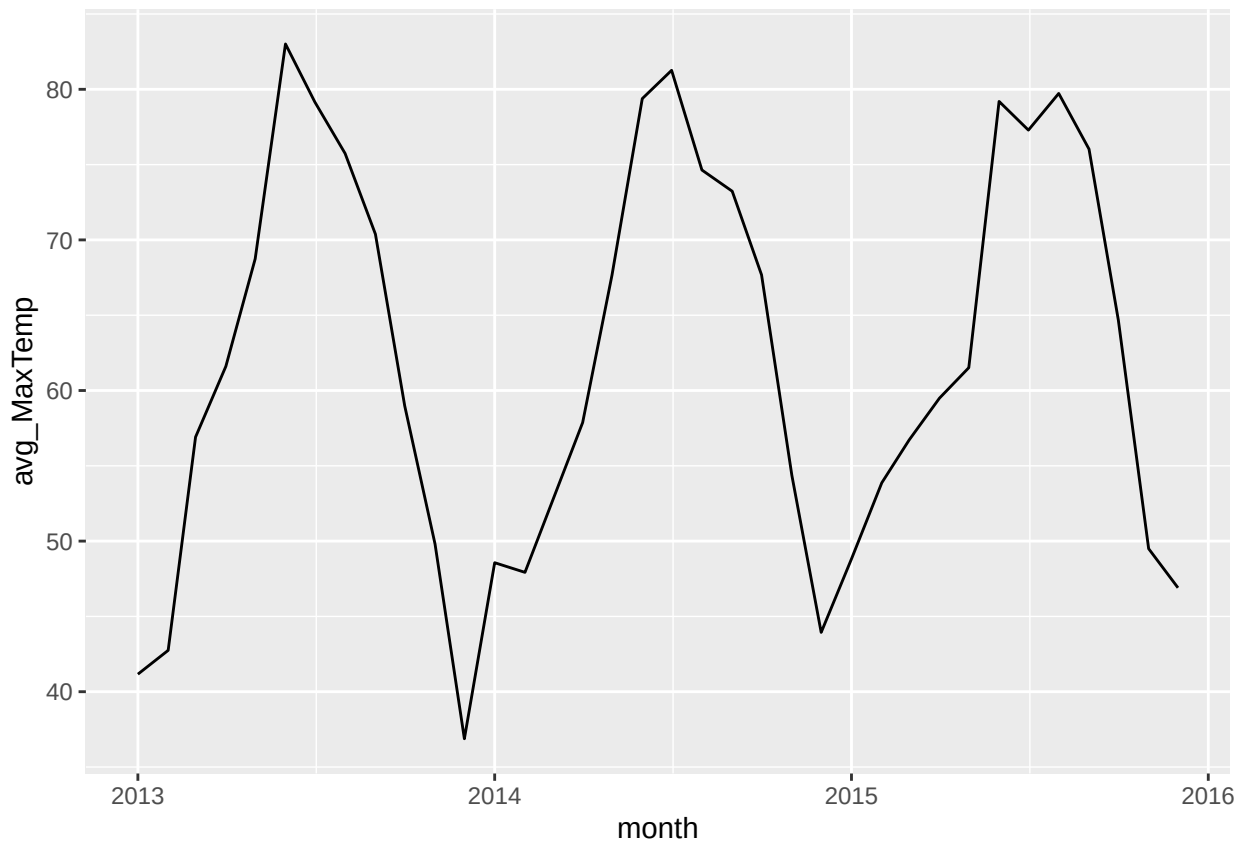
ggplot(temp_2005, aes(x = date, y = MaxTemp)) +
  geom_line()
```



b) Create a line graph that gives the monthly average maximum temperature for 2013 - 2015. *Again the x-axis should be the date and span 3 years.*

```
temp_monthly <-
  temperature_long %>%
  filter(year(date) >= 2013, year(date) <= 2015) %>%
  mutate(month = floor_date(date, unit = "month")) %>%
  group_by(month) %>%
  summarize(avg_MaxTemp = mean(MaxTemp, na.rm = TRUE), .groups = "drop") %>%
  arrange(month) %>%
  drop_na(avg_MaxTemp)

ggplot(temp_monthly, aes(x = month, y = avg_MaxTemp)) +
  geom_line()
```



Problem 4

The package `nycflights13` contains information about all the flights that arrived in or left from New York City in 2013. This package contains five data tables, but there are three data tables we will work with. The data table `flights` gives information about a particular flight, `airports` gives information about a particular airport, and `airlines` gives information about each airline. Create a table of all the flights on February 14th by Virgin America that has columns for the carrier, destination, departure time, and flight duration. Join this table with the airports information for the destination. Notice that because the column for the destination airport code doesn't match up between `flights` and `airports`, you'll have to use the `by=c("TableA.Col"="TableB.Col")` argument where you insert the correct names for TableA.Col and TableB.Col.

```
virgin_flights <-
  flights %>%
  filter(month == 2, day == 14, carrier == "VX") %>%
  select(carrier, dest, dep_time, air_time)

virgin_flights_airports <-
  virgin_flights %>%
  left_join(airports, by = c("dest" = "faa"))

head(virgin_flights_airports)
```

```
## # A tibble: 6 x 11
```

##	carrier	dest	dep_time	air_time	name	lat	lon	alt	tz	dst	tzone
##	<chr>	<chr>	<int>	<dbl>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>	<chr>
## 1	VX	LAX	706	347	Los Angel~	33.9	-118.	126	-8	A	Amer~
## 2	VX	SFO	732	344	San Franc~	37.6	-122.	13	-8	A	Amer~
## 3	VX	LAX	909	341	Los Angel~	33.9	-118.	126	-8	A	Amer~
## 4	VX	LAS	934	307	Mc Carran~	36.1	-115.	2141	-8	A	Amer~
## 5	VX	SFO	1029	351	San Franc~	37.6	-122.	13	-8	A	Amer~
## 6	VX	LAX	1317	349	Los Angel~	33.9	-118.	126	-8	A	Amer~

Problem 5

Data table joins are extremely common because effective database design almost always involves having multiple tables for different types of objects. To illustrate both table joins and the usefulness of multiple tables we will develop a set of data frames that will represent a credit card company's customer data base. We will have tables for Customers, Retailers, Cards, and Transactions. Below is code that will create and populate these tables.

```
Customers <- tribble(
  ~PersonID, ~Name, ~Street, ~City, ~State,
  1, 'Derek Sonderegger', '231 River Run', 'Flagstaff', 'AZ',
  2, 'Aubrey Sonderegger', '231 River Run', 'Flagstaff', 'AZ',
  3, 'Robert Buscaglia', '754 Forest Heights', 'Flagstaff', 'AZ',
  4, 'Roy St Laurent', '845 Elk View', 'Flagstaff', 'AZ')

Retailers <- tribble(
  ~RetailID, ~Name, ~Street, ~City, ~State,
  1, 'Kickstand Kafe', '719 N Humphreys St', 'Flagstaff', 'AZ',
  2, 'MartAnnes', '112 E Route 66', 'Flagstaff', 'AZ',
  3, 'REI', '323 S Windsor Ln', 'Flagstaff', 'AZ' )

Cards <- tribble(
  ~CardID, ~PersonID, ~Issue_DateTime, ~Exp_DateTime,
  '9876768717278723', 1, '2019-9-20 0:00:00', '2022-9-20 0:00:00',
  '5628927579821287', 2, '2019-9-20 0:00:00', '2022-9-20 0:00:00',
  '7295825498122734', 3, '2019-9-28 0:00:00', '2022-9-28 0:00:00',
  '8723768965231926', 4, '2019-9-30 0:00:00', '2022-9-30 0:00:00' )

Transactions <- tribble(
  ~CardID, ~RetailID, ~DateTime, ~Amount,
  '9876768717278723', 1, '2019-10-1 8:31:23', 5.68,
  '7295825498122734', 2, '2019-10-1 12:45:45', 25.67,
  '9876768717278723', 1, '2019-10-2 8:26:31', 5.68,
  '9876768717278723', 1, '2019-10-2 8:30:09', 9.23,
  '5628927579821287', 3, '2019-10-5 18:58:57', 68.54,
  '7295825498122734', 2, '2019-10-5 12:39:26', 31.84,
  '8723768965231926', 2, '2019-10-10 19:02:20', 42.83)

Cards <- Cards %>%
  mutate( Issue_DateTime = lubridate::ymd_hms(Issue_DateTime),
```



```

      Exp_DateTime = lubridate::ymd_hms(Exp_DateTime) )
Transactions <- Transactions %>%
  mutate( DateTime = lubridate::ymd_hms(DateTime))

```

a) Create a table that gives the credit card statement for Derek. It should give all the transactions, the amounts, and the store name. Write your code as if the only initial information you have is the customer's name. *Hint: Do a bunch of table joins, and then filter for the desired customer name. To be efficient, do the filtering first and then do the table joins.*

```

statement_derek <-
  Customers %>%
    filter(Name == "Derek Sonderegger") %>%
    left_join(Cards, by = "PersonID") %>%
    left_join(Transactions, by = "CardID") %>%
    left_join(Retailers, by = "RetailID") %>%
    select(Name = Name.x, DateTime, Amount, Store = Name.y)
statement_derek

```

```

## # A tibble: 3 x 4
##   Name      DateTime      Amount Store
##   <chr>      <dtm>      <dbl> <chr>
## 1 Derek Sonderegger 2019-10-01 08:31:23    5.68 Kickstand Kafe
## 2 Derek Sonderegger 2019-10-02 08:26:31    5.68 Kickstand Kafe
## 3 Derek Sonderegger 2019-10-02 08:30:09    9.23 Kickstand Kafe

```

b) Aubrey has lost her credit card on Oct 15, 2019. Close her credit card at 4:28:21 PM and issue her a new credit card in the `Cards` table. *Hint: Using the Aubrey's name, get necessary CardID and PersonID and save those as `cardID` and `personID`. Then update the `Cards` table row that corresponds to the `cardID` so that the expiration date is set to the time that the card is closed. Then insert a new row with the `personID` for Aubrey and a new `CardID` number that you make up.*

```

aubrey_info <-
  Customers %>%
    filter(Name == "Aubrey Sonderegger") %>%
    left_join(Cards, by = "PersonID") %>%
    select(PersonID, CardID)

cardID <- aubrey_info %>% pull(CardID)
personID <- aubrey_info %>% pull(PersonID)

Cards <- Cards %>%
  mutate(
    Exp_DateTime = if_else(
      CardID == cardID,
      lubridate::ymd_hms("2019-10-15 16:28:21"),
      Exp_DateTime)) %>%

  add_row(

```

```
CardID = "5628927579821288",
PersonID = personID,
Issue_DateTime = lubridate::ymd_hms("2019-10-15 16:28:21"),
Exp_DateTime = lubridate::ymd_hms("2022-10-15 16:28:21"))
```

Cards

```
## # A tibble: 5 x 4
##   CardID      PersonID Issue_DateTime      Exp_DateTime
##   <chr>      <dbl> <dtm>      <dtm>
## 1 9876768717278723      1 2019-09-20 00:00:00 2022-09-20 00:00:00
## 2 5628927579821287      2 2019-09-20 00:00:00 2019-10-15 16:28:21
## 3 7295825498122734      3 2019-09-28 00:00:00 2022-09-28 00:00:00
## 4 8723768965231926      4 2019-09-30 00:00:00 2022-09-30 00:00:00
## 5 5628927579821288      2 2019-10-15 16:28:21 2022-10-15 16:28:21
```

c) Aubrey is using her new card at Kickstand Kafe on Oct 16, 2019 at 2:30:21 PM for coffee with a charge of \$4.98. Generate a new transaction for this action. *Hint: create temporary variables `card`, `retailid`, `datetime`, and `amount` that contain the information for this transaction and then write your code to use those. This way in the next question you can just use the same code but modify the temporary variables. Alternatively, you could write a function that takes in these four values and manipulates the tables in the GLOBAL environment using the `<-` command to assign a result to a variable defined in the global environment. The reason this is OK is that in a real situation, these data would be stored in a database and we would expect the function to update that database.*

```
card <- Cards %>%
  filter(PersonID == 2) %>%
  arrange(desc(Issue_DateTime)) %>%
  slice(1) %>%
  pull(CardID)

retailid <- Retailers %>%
  filter(Name == "Kickstand Kafe") %>%
  pull(RetailID)

datetime <- lubridate::ymd_hms("2019-10-16 14:30:21")
amount <- 4.98

Transactions <- Transactions %>%
  add_row(
    CardID = card,
    RetailID = retailid,
    DateTime = datetime,
    Amount = amount)
Transactions
```

```
## # A tibble: 8 x 4
##   CardID      RetailID DateTime      Amount
```

	<chr>	<dbl>	<dtm>	<dbl>
## 1	9876768717278723	1	2019-10-01 08:31:23	5.68
## 2	7295825498122734	2	2019-10-01 12:45:45	25.7
## 3	9876768717278723	1	2019-10-02 08:26:31	5.68
## 4	9876768717278723	1	2019-10-02 08:30:09	9.23
## 5	5628927579821287	3	2019-10-05 18:58:57	68.5
## 6	7295825498122734	2	2019-10-05 12:39:26	31.8
## 7	8723768965231926	2	2019-10-10 19:02:20	42.8
## 8	5628927579821288	1	2019-10-16 14:30:21	4.98

d) On Oct 17, 2019, some nefarious person is trying to use her OLD credit card at REI. Make sure your code in part (c) first checks to see if the credit card is active before creating a new transaction. Using the same code, verify that the nefarious transaction at REI is denied. *Hint: your check ought to look something like this:*

I was a little confused with what the question was asking me I hope I understood it correctly

```
card <- "5628927579821287"
retailid <- 3
datetime <- ymd_hms("2019-10-17 16:28:21")
amount <- 4.98

# If the card is currently valid, this should return exactly 1 row.
Valid_Cards <- Cards %>%
  filter(CardID == card, Issue_DateTime <= datetime, datetime <= Exp_DateTime)

# If the transaction is valid, insert the transaction into the table
if( nrow(Valid_Cards) == 1){
  Transactions <- Transactions %>%
    add_row(
      CardID = card,
      RetailID = retailid,
      DateTime = datetime,
      Amount = amount
    )
}else{
  print('Card Denied')
}
```

```
## [1] "Card Denied"
```

e) Generate a table that gives the credit card statement for Aubrey. It should give all the transactions, amounts, and retailer name for both credit cards she had during this period.

```
statement_aubrey <- Customers %>%
  filter(Name == "Aubrey Sonderegger") %>%
  left_join(Cards, by = "PersonID") %>%
  left_join(Transactions, by = "CardID") %>%
  left_join(Retailers, by = "RetailID") %>%
```

```

filter(!is.na(DateTime)) %>%
arrange(DateTime) %>%
select(Name = Name.x, CardID, DateTime, Amount, Store = Name.y)
statement_aubrey

```

```

## # A tibble: 2 x 5
##   Name          CardID      DateTime      Amount Store
##   <chr>         <chr>      <dtm>      <dbl> <chr>
## 1 Aubrey Sonderegger 5628927579821287 2019-10-05 18:58:57 68.5 REI
## 2 Aubrey Sonderegger 5628927579821288 2019-10-16 14:30:21  4.98 Kickstand Kafe

```